Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29 June 2026
Team ID	
Project Name	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion & Extraction	Connect seamlessly to the raw source file **food_coded.csv**. Generate specialized Tableau Data Extracts (`.hyper`) to optimize row loading speeds. Parse and assign correct data types to user variables (e.g., categorical integers to text filters like `Gender` or `cuisine`).
FR-2	Student Wellness Analysis (Rahul's Persona)	Build scatter plots correlating breakfast consumption frequency against GPA scores. Develop heat maps analyzing late-night junk food consumption against daytime concentration levels. Provide clear visual highlights showing the academic impact of positive food choices.
FR-3	Stress & Emotional Eating Tracking (Priya's Persona)	Create visual categorizations isolating emotional eating triggers (`comfort_food_reasons_coded`). Provide stacked area charts mapping comfort food types against stressful campus periods (e.g., exams). Integrate a clean baseline visual comparing peer dietary behaviors.
FR-4	Campus Cafeteria Optimization Dashboard	Deliver comparison bar charts contrasting student healthy food desires vs. fast food purchases. Construct a dynamic pricing matrix evaluating student willingness to spend on campus meal combos. Generate automated alerts highlighting low-consumption nutritional categories (e.g., missing fruits/vegetables).
FR-5	Multi-Dimensional Filtering	Implement structural dashboard filters based on cultural cuisine choices (`cuisine`). Provide demographic cross-filtering features using variables like `Gender` and academic year. Ensure instantaneous, real-time recalculations whenever a dashboard filter is adjusted.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The user interface must be self-explanatory for students and non-technical stakeholders (like cafeteria personnel). Interactive filters must require no more than 2 clicks to surface desired segments.
NFR-2	**Security**	Dashboards must comply with campus data privacy standards. Individual student identity fields are masked/anonymized within **food_coded.csv**; only aggregated data trends must be made public.
NFR-3	**Reliability**	The application must render visual data with 100% computational accuracy, ensuring zero layout breakages or data mismatches during standard student filter combinations.
NFR-4	**Performance**	Dashboards must load fully in less than 3 seconds. Visual updates, cross-filtering requests, and parameter changes must respond instantly (<1 second latency).
NFR-5	**Availability**	The final Tableau dashboards must be hosted on Tableau Cloud or an institutional portal, achieving 99.9% uptime for 24/7 access by students and staff during the academic semester.
NFR-6	**Scalability**	The backend data pipeline must be constructed flexibly so that data files from future survey cycles can be appended directly to the existing data framework without breaking the charts.